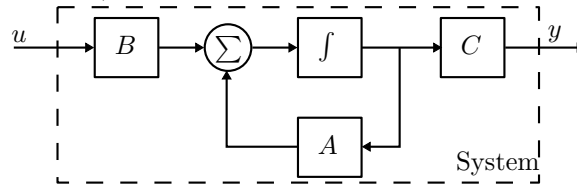


Test: Introducing Reference Signals

1. Complete the following diagram with a *full order observer*, a *state feedback*, and a *reference signal* (not integral action).



2. Given a system with poles at -1 , -2 , -3 , and -4 (the open-loop system has no zeros). Where should two zeros be placed to maximize the bandwidth of the system?
3. Determine if the following system is controllable and observable, by calculating its controllability matrix and observability matrix.

$$\begin{aligned}\dot{x} &= \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u \\ y &= \begin{bmatrix} 1 & 0 \end{bmatrix} x.\end{aligned}$$

4. Compute the zeros of the following system

$$\begin{aligned}\dot{x} &= \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \\ y &= \begin{bmatrix} 0 & 1 \end{bmatrix} x.\end{aligned}$$